

Further innovations

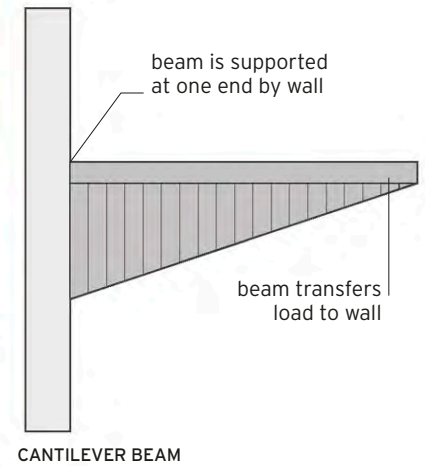
Steel, along with another piece of new technology, the elevator, was key to the most significant innovation in modern architecture: the skyscraper. It began in Chicago, with the 10-story Home Insurance Company Building of 1885, before the momentum shifted to New York, where the Manhattan Life Building hit 26 stories in 1889, followed by the Singer Building of 47 stories in 1907, and topping out with the 102-story Empire State Building (see p.43) in 1931.

As with height for buildings, steel facilitated longer spans for bridges. In 1874, the Eads Bridge over the Mississippi River at St. Louis was completed. A distance of 520 ft (158 m) between the piers made it the longest rigid span ever built at the time, which was achieved through the use of a steel arch. Nine years later, the monumental Brooklyn Bridge (see pp.26–27) opened in New York as the world's first major suspension bridge with steel cables, suspenders, and a deck. It was the first step toward future grand spans, such as the Golden Gate Bridge in San Francisco (see pp.46–47), with its main span of 4,200 ft (1,280 m), and the Akashi Kaikyo Bridge (see p.308) in Japan—the world's longest—with a main span of 6,532 ft (1,991 m).

Due to environmental concerns, **100 percent recyclable steel** is now being used to build many structures

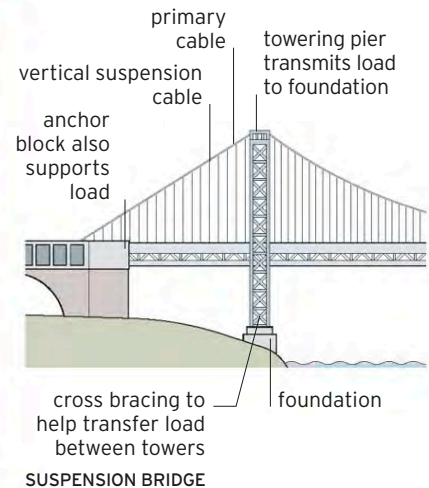
Redefining architecture

The industrial age also saw the reintroduction of concrete, a material used in Roman times but now revived in reinforced form. Engineers were now able to create a dome of a size that surpassed ancient Rome's Pantheon (see pp.104–105), with the Centennial Hall at Breslau, Germany (now Wrocław, Poland), in 1913. The facility to precast concrete in almost any shape freed architects' minds, resulting in the visionary church Notre Dame-du-Haut by Le Corbusier, the organic houses of Frank Lloyd Wright, and Jørn Utzon's sail-like Sydney Opera House (see pp.304–305). The trend continues, but not just with concrete. It is aided by computer modeling of new buildings, which now lean, twist, loom, hover, and of course climb even higher.



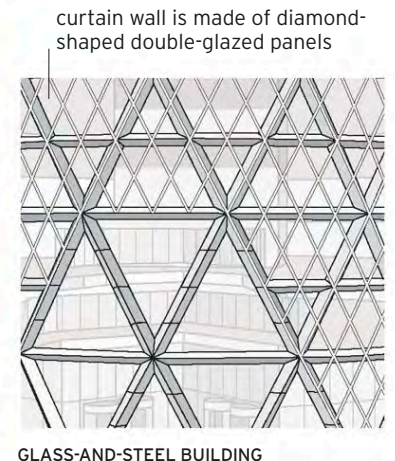
△ CANTILEVER

The cantilever, in which only one end of a structure is supported, has often been used in bridge structures. However, in the China Central Television (CCTV) Building (shown above), built in 2015 in Beijing, the whole upper portion of the tower is cantilevered for 245 ft (75 m).



△ SUSPENSION

As a material, steel works well in compression but also in tension, which is exploited in suspension bridges such as New York's George Washington Bridge. In such structures, the deck of the bridge is suspended from giant cables that are hung off towering piers.



△ NEW MATERIALS AND DESIGN TOOLS

By taking advantage of new materials and manufacturing techniques, architects can now design buildings with sinuous curves and irregular outlines, such as Frank Gehry's Weisman Art Museum in Minnesota. Many modern buildings are enclosed by an outer curtain wall, often made of metal and glass.